# Land8Fire — Digital Twin

## Wildfire Physics & Environmental Parameters Presentation Briefing

**Real-World 3D Twin**
Rothermel Physics Engine
Presentation & Defense Guide

## 1. System Overview & Digital Twin Architecture

**Land8Fire** is a high-performance 3D geospatial Wildfire Digital Twin engineered with **React 18**, **TypeScript**, and **CesiumJS**. It couples real-time meteorological observations with an authoritative **Rothermel (1972) surface fire spread physics engine** and a **multispectral deep learning U-Net model** for burned area segmentation.

### Role of the Environmental Parameters Panel

The **Environmental Parameters Panel** is not merely an informational dashboard; it provides the **thermodynamic and fluid-dynamic boundary conditions** directly consumed by the fire propagation differential equations. Every parameter dynamically modulates the rate and geometry of the flame front.

## 2. Mathematical & Physical Foundation: Rothermel Model

Wildfire spread is fundamentally governed by **heat transfer balance**: the rate at which heat is released in the reaction zone versus the heat required to ignite adjacent unburned fuel ($Q_{ig}$).

```
// Directional Rate of Spread (ROS) across 36 sectors (Delta Theta = 10 deg) ROS(θ) = ROS_0 × Burnability × η_M × TempBoost × φ_wind(θ) × φ_slope(θ) // Where: ROS_0 = Baseline forward spread rate of the fuel bed (~2.5 m/min) Burnability = ESA WorldCover / Fuel model coefficient (0.0 to 1.0) η_M = Rothermel Fuel Moisture Damping Coefficient TempBoost = Ambient preheating factor: max(0.5, 1 + (T_ambient - 20) / 40) φ_wind(θ) = Directional wind modifier: max(0.1, 1 + c_w × U_wind × cos(θ - θ_wind)) φ_slope(θ) = Terrain slope gradient modifier: max(0.2, 1 + c_s × tan(slope) × cos(θ - aspect))
```

## 3. Environmental Parameters: Physics & Propagation Impact

| Parameter | Metric / Units | Physical Mechanism & Impact on Fire Spread |
|---|---|---|
| **Temperature** | DEGREE CELSIUS (°C) | **Preheating of Fine Fuels:** Cellulosic forest fuels must reach pyrolysis temperature (~300°C–320°C) to ignite. Higher ambient temperature reduces the enthalpy deficit: $Q_{ig} = C_p \times (T_{ignite} - T_{ambient}) + M_f \times \Delta H_{vap}$. Less combustion heat is wasted preheating the fuel bed, accelerating forward spread. |
| **Relative Humidity** | PERCENTAGE (%) | **Vapor Pressure Deficit (VPD):** Governs atmospheric drying suction. Low RH (<30%) rapidly extracts water from dead leaves and needles. High RH drives moisture absorption, dampening ignition potential. |
| **1-Hour Dead Fuel Moisture (EMC)** | PERCENTAGE (%)<br>Simard / Rothermel | **Latent Heat of Vaporization Barrier:** Water absorbs *2,260 kJ/kg* to vaporize. As fuel moisture rises, Rothermel damping factor $\eta_M$ drops precipitously: $\eta_M = 1 - 2.59(M_f/M_x) + 5.11(M_f/M_x)^2 - 3.52(M_f/M_x)^3$. At **$M_x$ = 30% (Moisture of Extinction)**, $\eta_M = 0$ and fire spread stops entirely. |
| **Wind Speed & Direction** | KM/H & BEARING (°)<br>Compass Vector | **Forced Convection & Flame Tilting:** Wind tilts the buoyant flame geometry forward over unburned vegetation, exponentially increasing radiant flux and preheating. Transforms circular spread into an elongated **ellipse** along the wind vector. |
| **Precipitation** | MILLIMETERS (MM) | **Direct Surface Quenching:** Liquid rain coats fuel surfaces, driving effective moisture beyond extinction ($e^{-0.6 \times precip}$ damping), arresting the flame front. |
| **Surface Pressure** | HPA / BAROMETRIC | **Atmospheric Stability & Winds:** High-pressure systems (>1013 hPa) generate sinking air (subsidence), resulting in compression warming, low humidity, and high wildfire danger. |

## 4. Ember Spotting & River Crossing Mechanics

A critical capability of the Land8Fire Digital Twin is distinguishing between **direct surface flame propagation** and **aerodynamic ember spotting (river crossing)**:

### 1. Water Boundary Clamping (ESA WorldCover 10m)

As each of the 36 sector rays expands outward, the simulation queries high-resolution 10m land-cover classification. When a sector ray reaches a water body (river, lake, wetland), direct surface fire spread is **strictly clamped at the water's edge** (0 m/min).

### 2. Downwind Ember Ballistics & Spot Fire Ignition

In strong winds, burning firebrands (embers) are lofted into the convection column and transported downwind. Our model calculates the maximum aerodynamic jump reach:

$$Max_Spotting_Distance \approx U_{wind} \times 35 \times \cos(\theta - \theta_{wind}) \times (1.4 - RH / 100)$$

If the opposite bank of the river is burnable forest land and lies within this reach, the Digital Twin spawns a **secondary spot fire** with its own expanding elliptical perimeter, accurately modeling real-world river crossings.

## 5. Presentation Script & Speaking Points for the Evaluation Panel

Deliver these concise, high-impact speaking points when presenting the Environmental Parameters panel:

> *"Respected Panel Members:*
>
> *In our Digital Twin architecture, the **Environmental Parameters Panel** is not a static telemetry widget; it acts as the **thermodynamic boundary condition** that drives our real-time physics engine.*
>
> ***1. Atmospheric Coupling to Fuel Chemistry:** Rather than guessing spread rates, our engine dynamically computes the **1-Hour Dead Fuel Equilibrium Moisture Content (EMC)** using the Simard-Rothermel equations. Water requires 2.26 MJ/kg to vaporize. When our panel shows relative humidity dropping and fuel moisture falling below 8%, the physics engine reduces the moisture damping factor ($\eta_M$), triggering rapid, explosive spread.*
>
> ***2. Directional Vectoring & Flame Tilting:** Wildfires rarely expand in simple circles. Our engine projects the wind speed and compass bearing across **36 radial sectors**. Wind tilts the flame front forward, dramatically amplifying radiative and convective heat transfer downwind to produce realistic elliptical burn perimeters.*
>
> ***3. River Crossing & Ember Spotting:** While surface flames are halted at riverbanks using ESA WorldCover 10m data, high winds and dry air trigger our **ballistic spotting model**, enabling embers to bridge river gaps and ignite secondary spot fires on opposite banks.*
>
> *Every parameter on this panel directly modulates the differential equations governing the 3D fire perimeter on the Cesium globe."*

## 6. Anticipated Panel Questions & Authoritative Answers

**Q1: Why use 36 radial sectors instead of a standard 2D grid cellular automata?**
*Answer:* Cellular automata suffer from grid bias (distorting diagonal spread). The 36-sector vector-front formulation preserves true continuous compass azimuths, allows precise ellipsoidal modeling under shifting wind vectors, and renders natively at 60 FPS in Cesium 3D with zero rasterization latency.

**Q2: How does the Simard-Rothermel EMC differ from Live Canopy Moisture?**
*Answer:* Live fuel moisture changes slowly across seasons (deciduous vs. evergreen), whereas 1-Hour Dead Fuel Moisture responds within 60 minutes to hourly ambient temperature and relative humidity. It is the single most volatile factor in determining flashover and surface fire acceleration.

**Q3: How is this validated against real-world wildfire data?**
*Answer:* The Digital Twin includes the Copernicus EMSR239 benchmark disaster event and real Amazon wildfire observations, comparing simulated perimeters against Copernicus Emergency Management ground-truth delimitation maps and Sentinel-2 multispectral U-Net predictions.